

Column Selection for the Airline On Time Performance Database

Purpose and decision

We selected the U.S. Department of Transportation Bureau of Transportation Statistics Reporting Carrier On Time Performance data for January through June 2026. The original monthly export contains 53 source columns. The six monthly files will be combined into one staging dataset, and the final MySQL database will use the fields needed to analyze airline reliability, delays, cancellations, diversions, airports, routes, and monthly trends.

Important: We are not deleting information from the downloaded source files. We will keep all 53 source columns in the raw staging table, then map the required fields into normalized tables. This preserves reproducibility while avoiding unnecessary attributes in the final relational design.

Why these columns were selected

The selected fields were chosen because each supports a project question, a database relationship, or a required performance measure. Together they provide time, categorical, and numeric data and support joins, filtering, aggregation, subqueries, CTEs, and window functions.

Field group	Raw columns retained	Reason for retention
Date and time period	YEAR, QUARTER, MONTH, DAY_OF_MONTH, DAY_OF_WEEK, FL_DATE	Supports monthly trends, date analysis, and the date dimension.
Carrier and flight	OP_UNIQUE_CARRIER, OP_CARRIER_AIRLINE_ID, TAIL_NUM, OP_CARRIER_FL_NUM	Identifies the reporting/operating carrier, aircraft, and flight number.
Origin airport	ORIGIN_AIRPORT_ID, ORIGIN, ORIGIN_CITY_NAME, ORIGIN_STATE_ABR, ORIGIN_STATE_NM, ORIGIN_WAC	Supports airport relationships, state comparisons, and origin performance.
Destination airport	DEST_AIRPORT_ID, DEST, DEST_CITY_NAME, DEST_STATE_ABR, DEST_STATE_NM, DEST_WAC	Supports route relationships, destination analysis, and route-level queries.
Departure performance	CRS_DEP_TIME, DEP_TIME, DEP_DELAY, DEP_DELAY_NEW, DEP_DEL15, TAXI_OUT, WHEELS_OFF	Measures scheduled versus actual departure performance and departure delay status.
Arrival performance	WHEELS_ON, TAXI_IN, CRS_ARR_TIME, ARR_TIME, ARR_DELAY, ARR_DELAY_NEW, ARR_DEL15	Measures scheduled versus actual arrival performance and arrival delay status.

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Destination airport	DEST_AIRPORT_ID, DEST, DEST_CITY_NAME, DEST_STATE_ABR, DEST_STATE_NM, DEST_WAC	Supports route relationships, destination analysis, and route-level queries.
Flight status	CANCELLED, CANCELLATION_CODE, DIVERTED	Supports cancellation and diversion analysis.
Duration and distance	CRS_ELAPSED_TIME, ACTUAL_ELAPSED_TIME, AIR_TIME, DISTANCE	Supports route distance, schedule accuracy, and flight duration analysis.
Delay causes	CARRIER_DELAY, WEATHER_DELAY, NAS_DELAY, SECURITY_DELAY, LATE_AIRCRAFT_DELAY	Identifies and compares the causes of delays by airline, airport, route, and month.
Diversion summary	DivAirportLandings, DivReachedDest, DivActualElapsedTime, DivArrDelay, DivDistance	Supports diversion counts, diversion outcomes, diverted-flight elapsed time, arrival delay, and diversion distance.

Final 53-column project scope:

Date and time: YEAR, QUARTER, MONTH, DAY_OF_MONTH, DAY_OF_WEEK, FL_DATE

Carrier and flight: OP_UNIQUE_CARRIER, OP_CARRIER_AIRLINE_ID, TAIL_NUM, OP_CARRIER_FL_NUM

Origin and destination: ORIGIN_AIRPORT_ID, ORIGIN, ORIGIN_CITY_NAME, ORIGIN_STATE_ABR, ORIGIN_STATE_NM, ORIGIN_WAC, DEST_AIRPORT_ID, DEST, DEST_CITY_NAME, DEST_STATE_ABR, DEST_STATE_NM, DEST_WAC

Departure and arrival performance: CRS_DEP_TIME, DEP_TIME, DEP_DELAY, DEP_DELAY_NEW, DEP_DEL15, TAXI_OUT, WHEELS_OFF, WHEELS_ON, TAXI_IN, CRS_ARR_TIME, ARR_TIME, ARR_DELAY, ARR_DELAY_NEW, ARR_DEL15

Status, duration, distance, and delay causes: CANCELLED, CANCELLATION_CODE, DIVERTED, CRS_ELAPSED_TIME, ACTUAL_ELAPSED_TIME, AIR_TIME, DISTANCE, CARRIER_DELAY, WEATHER_DELAY, NAS_DELAY, SECURITY_DELAY, LATE_AIRCRAFT_DELAY

Added diversion summary: DivAirportLandings, DivReachedDest, DivActualElapsedTime, DivArrDelay, DivDistance

Why other BTS fields are not required

The BTS table contains additional fields that are useful for other studies but are outside this project's objectives. Excluding them from the final normalized schema reduces unnecessary storage, simplifies the ER diagram, and keeps the SQL queries focused. The original raw files remain available if a future query requires one of these fields.

Category	Examples	Reason not required in final schema
Summary percentages	OntimeArrivalPct, OntimeDeparturePct	We calculate on-time percentages from flight-level delay indicators, so storing summaries would duplicate results.
Redundant identifiers	Carrier codes or IDs not present in the selected export; duplicate code variants	The selected carrier identifier and DOT airline ID are sufficient for this six-month analysis.
Extra airport identifiers	Airport sequence IDs, city-market IDs, state FIPS, and similar alternate keys	The selected airport IDs, airport codes, city, state, and WAC fields support the required airport and route analysis.
Pre-grouped time fields	Departure/arrival time blocks and delay groups	Time blocks and groups can be derived from the scheduled times and delay values when needed.
Detailed diversion fields	Div1 through Div5 airport, wheels-on, ground-time, wheels-off, and tail fields	The project needs to count and compare diverted flights, not reconstruct every diversion stop.
Unrelated operational details	Fields outside delays, cancellations, diversions, airports, routes, dates, and flight performance	They do not support the project's stated research questions or required SQL techniques.

Data quality and preparation decisions

- Keep the six original monthly CSV files unchanged as source evidence.
- Combine the files into one raw staging table after confirming identical headers.
- Convert blank values to NULL rather than treating missing actual times as zero.
- Preserve negative delay values because they represent early departures or arrivals.
- Convert FL_DATE to a MySQL DATE value during cleaning; the source includes a time portion at midnight.
- Keep scheduled time fields as raw values during staging because leading zeros are meaningful, then convert them consistently for analysis.
- Convert CANCELLED and DIVERTED into clear Boolean or indicator values in the final database.
- Load parent tables before child tables so primary-key and foreign-key relationships can be enforced.